

Conjecture 00000002158 is false:
the chromatic threshold of C_5 -free graphs is below $1/5$

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1 The conjecture and its reading

Statement. *Definition: The chromatic threshold is the minimum-degree/ $|V|$ threshold keeping bounded chromatic number for H -free graphs. Conjecture: The chromatic threshold for C_5 -free graphs is $1/5$; the determination of the threshold follows from a balancing analysis of the overlap structure.*

The second clause (“follows from a balancing analysis”) is not a mathematical assertion; the statement is a conjunction whose first conjunct, $\delta_\chi(C_5) = 1/5$, we refute.

Definition 1 (Allen, Böttcher, Griffiths, Kohayakawa, Morris, arXiv:1108.1746). “The chromatic threshold $\delta_\chi(H)$ of a graph H is the infimum of $d > 0$ such that there exists $C = C(H, d)$ for which every H -free graph G with minimum degree at least $d|G|$ satisfies $\chi(G) < C$.”

Conventions (all as in the Lean file).

- Graphs are finite simple graphs; every finite graph is isomorphic to one on $\{0, \dots, n-1\}$, so we quantify over all $n \in \mathbb{N}$ (including $n = 0$) and all $G : \text{SimpleGraph (Fin n)}$.
- C_5 is Mathlib’s `cycleGraph 5` on $\mathbb{Z}/5$. “ H -free” means Mathlib’s `H.Free G`: G has no subgraph (not necessarily induced) isomorphic to H , i.e. no injective homomorphism $H \rightarrow G$.
- $\delta(G)$ is Mathlib’s `G.minDegree` (equal to 0 on the empty vertex set).
- “ $\chi(G) < C$ ” is expressed as `G.Colorable K` ($\chi(G) \leq K$) with $K = C - 1$; this gives the same set of admissible d .

In Lean:

```
def admissible {k : ℕ} (H : SimpleGraph (Fin k)) : Set ℝ :=
  {d | 0 < d /\ exists K : ℕ, forall (n : ℕ) (G : SimpleGraph (Fin n))
    [DecidableRel G.Adj], H.Free G -> d * n <= G.minDegree -> G.Colorable K}
noncomputable def chromaticThreshold H : ℝ := sInf (admissible H)
```

The set is bounded below by 0, so its infimum is the usual one.

2 Main result

Theorem 2. $d = 1/6$ is admissible for C_5 with $K = 264$. Hence $\delta_\chi(C_5) \leq 1/6 < 1/5$; in particular $\delta_\chi(C_5) \neq 1/5$.

(It is known that in fact $\delta_\chi(C_5) = 0$, by Thomassen; we only need the elementary bound.)

Throughout, G is a C_5 -free graph on n vertices with $6\delta(G) \geq n$, $N(v)$ is the neighbourhood of v and $\text{cn}(a, b) = N(a) \cap N(b)$.

Lemma 3 (five vertices give a C_5 ; `cycle5_1e`). *If $a \sim b \sim c \sim d \sim e \sim a$ and $a \neq c$, $a \neq d$, $b \neq d$, $b \neq e$, $c \neq e$, then $C_5 \subseteq G$.*

Proof. Adjacent vertices are distinct, so all five are distinct, and $i \mapsto (a, b, c, d, e)_i$ is an injective homomorphism from C_5 . $\square$

Lemma 4 (`not_free_of_class`). *Let $u \sim u'$ with $u \neq s \neq u'$ and $|\text{cn}(u, s)| \geq 3$, $|\text{cn}(u', s)| \geq 3$. Then G contains a C_5 .*

Proof. Pick $w \in \text{cn}(u, s) \setminus \{u'\}$ and then $w' \in \text{cn}(u', s) \setminus \{u, w\}$ (possible by the cardinality bounds). Then $s \sim w \sim u \sim u' \sim w' \sim s$, and $s \neq u$, $s \neq u'$, $w \neq u'$, $w \neq w'$, $u \neq w'$, so the previous lemma applies. $\square$

Call a vertex set T *good* if $|\text{cn}(a, b)| \leq 2$ for all distinct $a, b \in T$ (`Good`).

Lemma 5 (private neighbourhoods; `card_mul_minDegree_1e`). *If T is good, then $|T|\delta(G) \leq n + |T| \cdot 2(|T| - 1)$.*

Proof. For $s \in T$ let $P(s) = N(s) \setminus \bigcup_{t \in T \setminus \{s\}} N(t)$. Then $\delta \leq |N(s)| = |P(s)| + |\bigcup_{t \neq s} \text{cn}(s, t)| \leq |P(s)| + 2(|T| - 1)$. The sets $P(s)$, $s \in T$, are pairwise disjoint (a vertex of $P(s)$ lies in $N(s)$, hence not in $P(s'')$ for $s'' \neq s$), so $\sum_{s \in T} |P(s)| \leq n$. Summing over $s \in T$ gives the claim. $\square$

Theorem 6 (`colorable22`). *If G is C_5 -free, $n \leq 6\delta(G)$ and $n > 264$, then G is 22-colourable.*

Proof. Let S be a good set of maximum cardinality (it exists: $\emptyset$ is good and V is finite). *Witnesses.* If $v \notin S$, then $S \cup \{v\}$ is not good by maximality, so some $s \in S$ has $|\text{cn}(v, s)| \geq 3$. *Size.* If $|S| \geq 12$, take a 12-element $T \subseteq S$ (good). Lemma 5 gives $12\delta \leq n + 264$, while $12\delta \geq 2n$; so $n \leq 264$, a contradiction. Thus $|S| \leq 11$. *Colouring.* Colour $s \in S$ by (true, s) and $v \notin S$ by (false, s_v) where $s_v \in S$ is a witness. Adjacent vertices in S get different second coordinates; a vertex in S and one outside get different first coordinates; two adjacent $u, v \notin S$ with $s_u = s_v = s$ would satisfy Lemma 4 ($u, v \neq s$ since $s \in S$), contradicting C_5 -freeness. The colour set $\{\text{true}, \text{false}\} \times S$ has $2|S| \leq 22$ elements. $\square$

Proof of Theorem 2 (`one_sixth_admissible`). Let G on n vertices be C_5 -free with $\frac{1}{6}n \leq \delta(G)$. If $n \leq 264$, colour each vertex by itself: G is n -, hence 264-colourable. Otherwise G is 22-colourable, hence 264-colourable. So $1/6 \in \text{admissible}(\text{cycleGraph } 5)$, the infimum is at most $1/6$ (`chromaticThreshold_cycle5_1e`), and $1/6 < 1/5$ (`chromaticThreshold_cycle5_ne`). $\square$

3 The induced reading

If “ C_5 -free” were read as “no *induced* C_5 ” (Mathlib’s `IsIndContained`, an embedding $C_5 \hookrightarrow G$ preserving adjacency and non-adjacency), with the same definition otherwise (`admissibleInd`, `chromaticThresholdInd`), the threshold is exactly 1:

- K_n contains no induced C_5 : an embedding f would make $f(0) \neq f(2)$ adjacent in K_n , while $0, 2$ are non-adjacent in C_5 (`top_not_indContains`).

- If $d < 1$ were admissible with constant K , choose $m > 1/(1 - d)$ and $n = m + K + 1$; then $\delta(K_n) = n - 1 \geq dn$, so K_n would be K -colourable, but $\chi(K_n) = n > K$ (`one_le_of_mem_admissibleInd`).
- $d = 1$ is admissible with $K = 0$: for $n \geq 1$, $\delta(G) \leq n - 1 < n$, so the degree hypothesis fails; for $n = 0$ the empty graph is 0-colourable (`one_mem_admissibleInd`).

Hence $\delta_\chi^{\text{ind}}(C_5) = 1 \neq 1/5$ (`chromaticThresholdInd_cycle5`).

4 Lean formalization and axiom audit

File `lean/Conjecture2158/Basic.lean` (246 lines, only import `Mathlib`, Lean/Mathlib v4.33.1), namespace `C2158`. Main theorems:

```
theorem chromaticThreshold_cycle5_ne :
  chromaticThreshold (cycleGraph 5) < 1 / 5 /\
  chromaticThreshold (cycleGraph 5) != 1 / 5
theorem chromaticThresholdInd_cycle5 :
  chromaticThresholdInd (cycleGraph 5) = 1 /\
  chromaticThresholdInd (cycleGraph 5) != 1 / 5
```

`#print axioms for chromaticThreshold_cycle5_ne, chromaticThresholdInd_cycle5 and one_sixth_admissible: [propext, Classical.choice, Quot.sound]. No sorry, no native_decide, no new axioms.`

Scope. We refute exactly two readings: the standard (non-induced subgraph) reading of ABGKM, and the induced-subgraph reading. We do not treat other threshold notions, e.g. the homomorphism threshold, for which the value $1/5$ appears for $\{C_3, C_5\}$ -free graphs (Letzter–Snyder); the statement’s own definition is in terms of bounded chromatic number, which is the chromatic threshold.

Source

P. Allen, J. Böttcher, S. Griffiths, Y. Kohayakawa, R. Morris, *The chromatic thresholds of graphs*, Adv. Math. 235 (2013); <https://arxiv.org/abs/1108.1746>.